

Communication Networks Laboratory

The University of Kansas EECS 780

Introduction to Protocol Analysis with Wireshark

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<https://www.ittc.ku.edu/~jpgs/courses/nets>

EECS Laboratories

Outline

- WS.0 EECS 780 laboratory outline
- WS.1 Motivation and overview
- WS.2 Wireshark installation and use
- WS.3 Protocol analysis examples
- WS.4 Getting started

EECS 780 Laboratories

Semester Outline

- Wireshark labs
 - throughout semester, intuitive, based on textbook
- Wiki and web authoring
 - requires KU, or ITTC account
- Socket programming
 - relatively simple lab to demonstrate socket concepts
- Network simulation
 - lab to introduce network simulation

Protocol Analysis with Wireshark

Motivation and Overview

- WS.0 EECS 780 laboratory outline
- WS.1 Motivation and overview**
- WS.2 Wireshark installation and use
- WS.3 Protocol analysis examples
- WS.4 Getting started

Motivation and Overview

Introduction

- Wireshark is a *network protocol analyzer*
 - captures network packets
 - displays packet data in details
 - www.wireshark.org
- First released in 1998 by Gerald Combs as Ethereal
 - many contributors around the world
- Open source and free software
- Graphical alternative to tcpdump

Motivation and Overview

Purpose

- Powerful tool for
 - troubleshooting network problems
 - examining security problems
 - debugging protocol implementations
 - learning network protocol internals
- Used in industry and academia

Protocol Analysis with Wireshark

Wireshark Installation and Use

- WS.0 EECS 780 laboratory outline
- WS.1 Motivation and overview
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Wireshark Installation

Highlights

- Wireshark can be installed on various platforms
 - UNIX, MS, Linux, Mac OS, etc.
- Most recent release is v.2.6.3, 29 August 2018
- System requirements
 - section 1.2 at http://www.wireshark.org/docs/wsug_html/
 - rule of thumb: fast CPU, more memory is better
- FAQs and Wiki pages provide more information

Wireshark Installation

Overview

- Installation of Wireshark requires
 - downloading the relevant package
 - building the source into binary if the source is downloaded
 - install binaries to their destinations
 - section 2 provides detailed installation instructions
http://www.wireshark.org/docs/wsug_html/
- Windows installation includes WinPcap
 - packet capture library (also needed for tcpdump)
- Installation easy and intuitive

Wireshark Usage

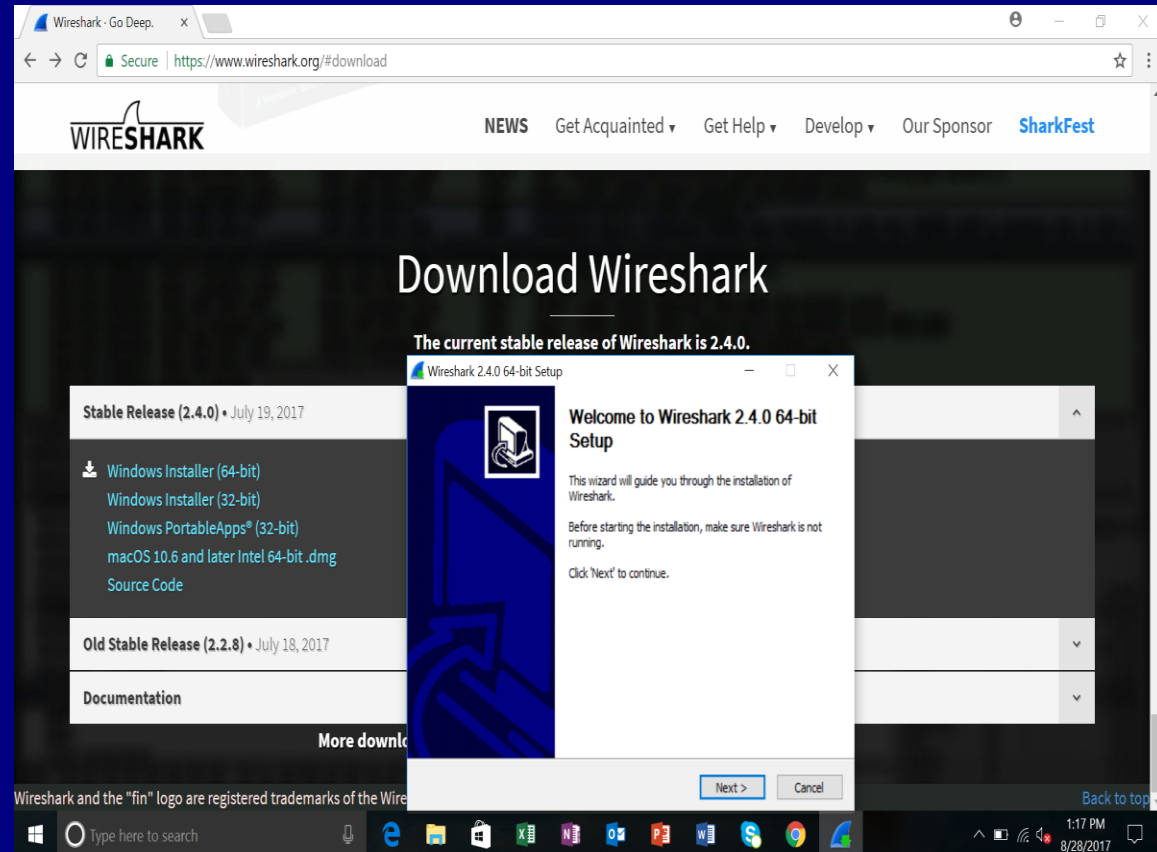
Windows 10 Installation₁

Go to
wireshark.org

Click on
*Download
Wireshark*

Save and run
the executable
.exe file

Installation
wizard is
intuitive



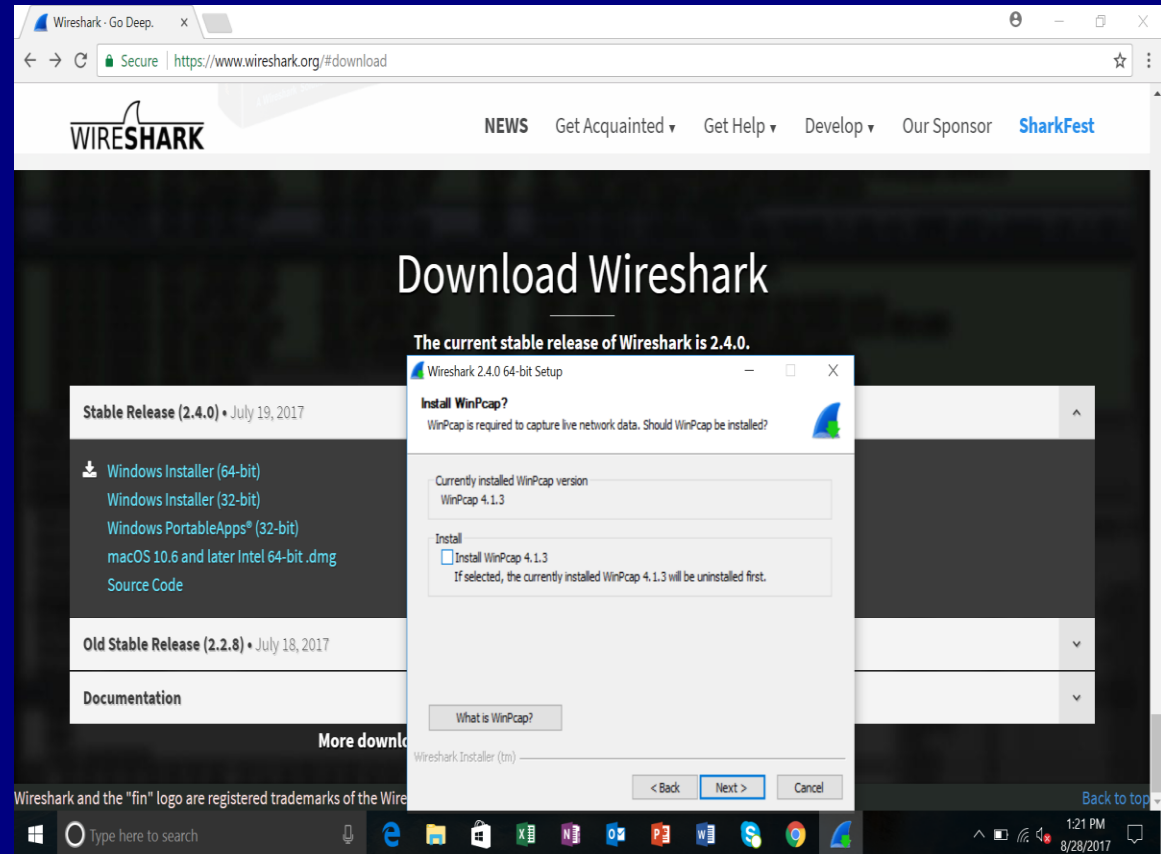
Wireshark Usage

Windows 10 Installation₂

pcap library is required to capture low-level network messages

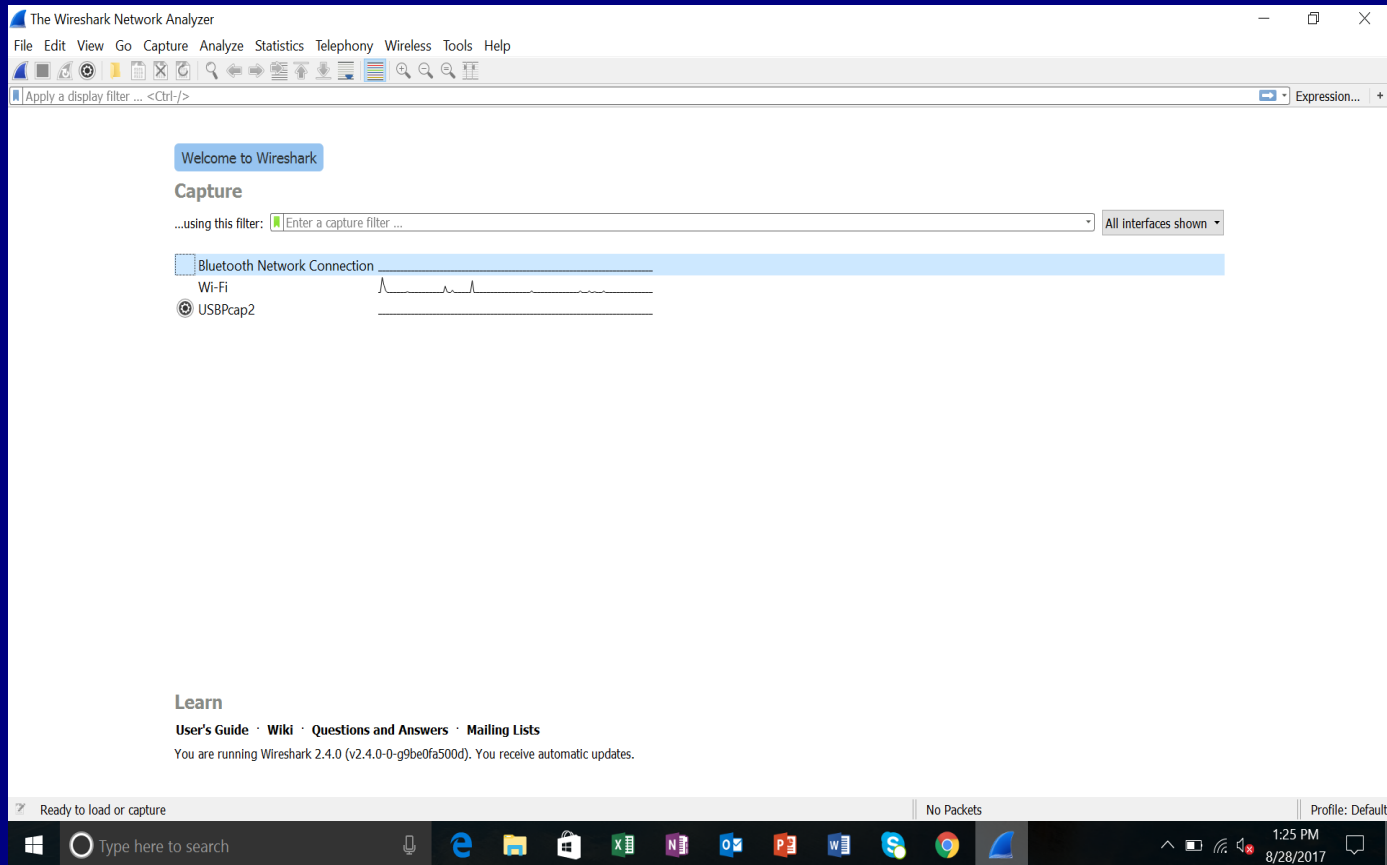
WinPcap for Windows,
libpcap for UNIX/Linux

Latest
WinPcap
release 4.1.3



Wireshark Installation

Windows 10 Installation₃



Wireshark Usage

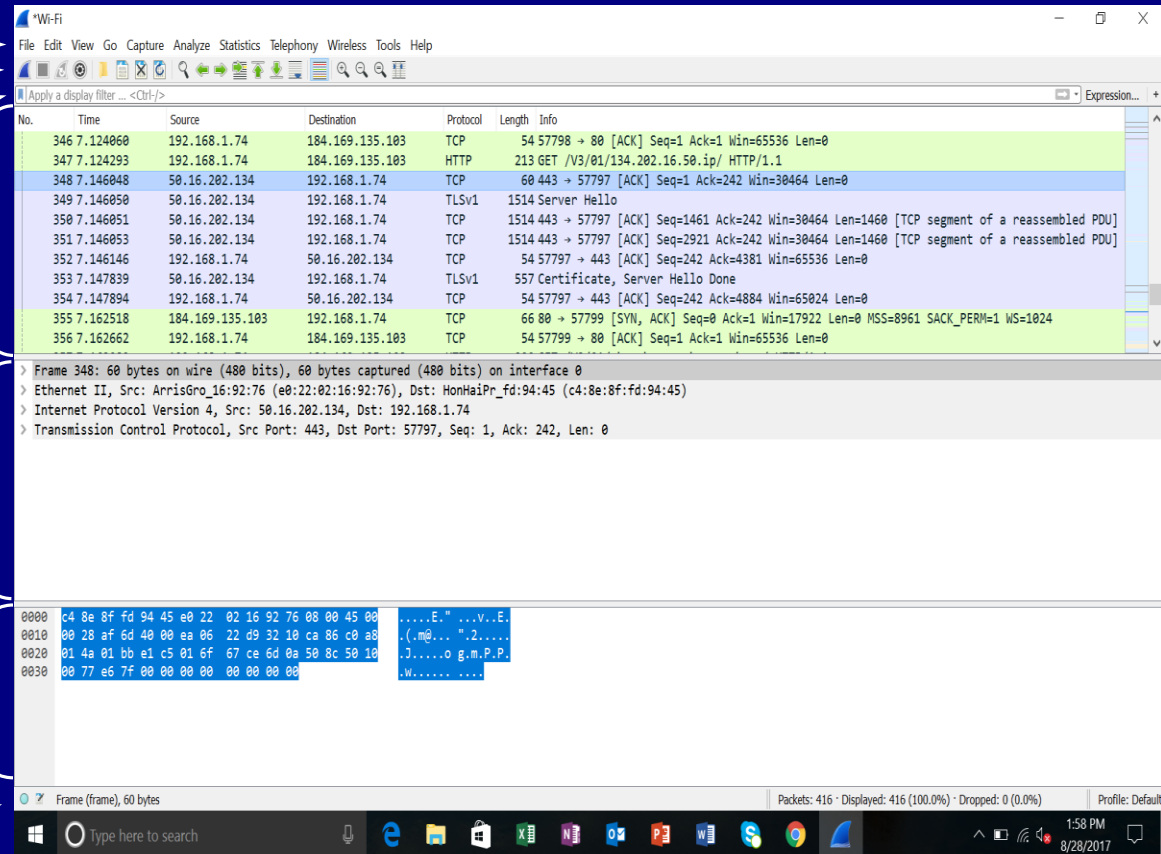
Main Features

- Capture live traffic
 - data can be captured on wired or wireless medium
 - numerous protocols can be captured and analyzed
- Display packet in details
- Open files containing packet data captured
 - from other programs (tcpdump/WinDump)
- Filtering is essential when dealing with lots of packets
 - filters can be applied on protocols, fields, values, etc.
 - filtering while capturing packets is possible

Wireshark GUI

Main Window

menu
main toolbar
filter field
packet list
pane
packet details
pane
packet bytes
pane
status bar



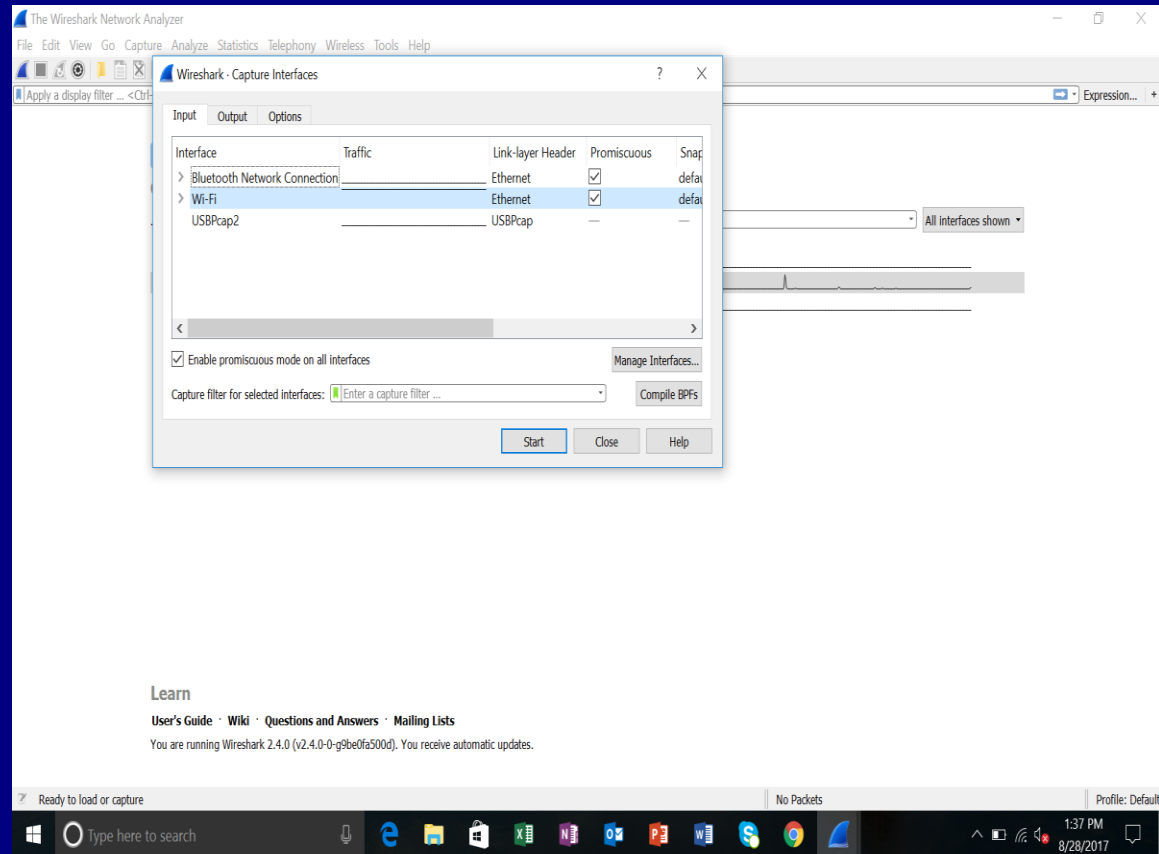
Wireshark Usage

Starting Capture

To capture:
go to Capture
menu and
select
Options...

Start
capturing on
interface that
has IP address

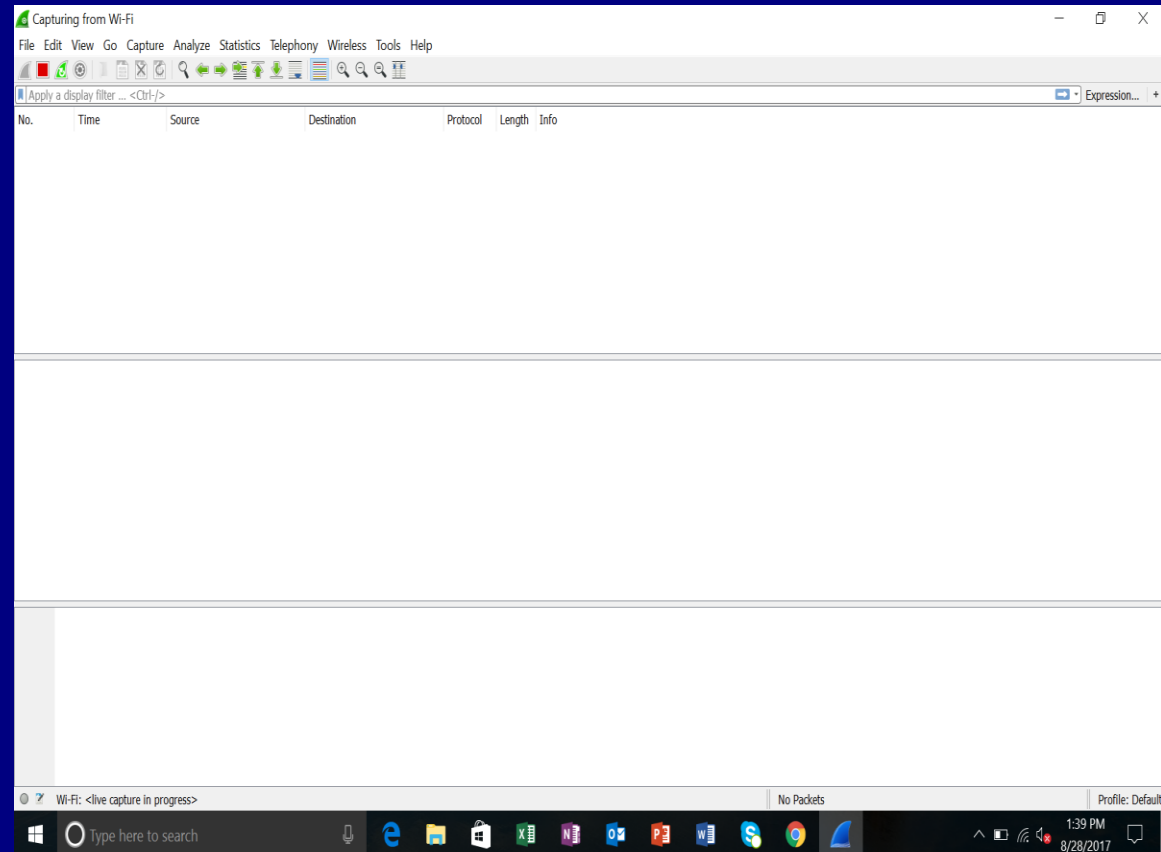
Other ways of
capturing
possible



Wireshark Usage

Capturing₁

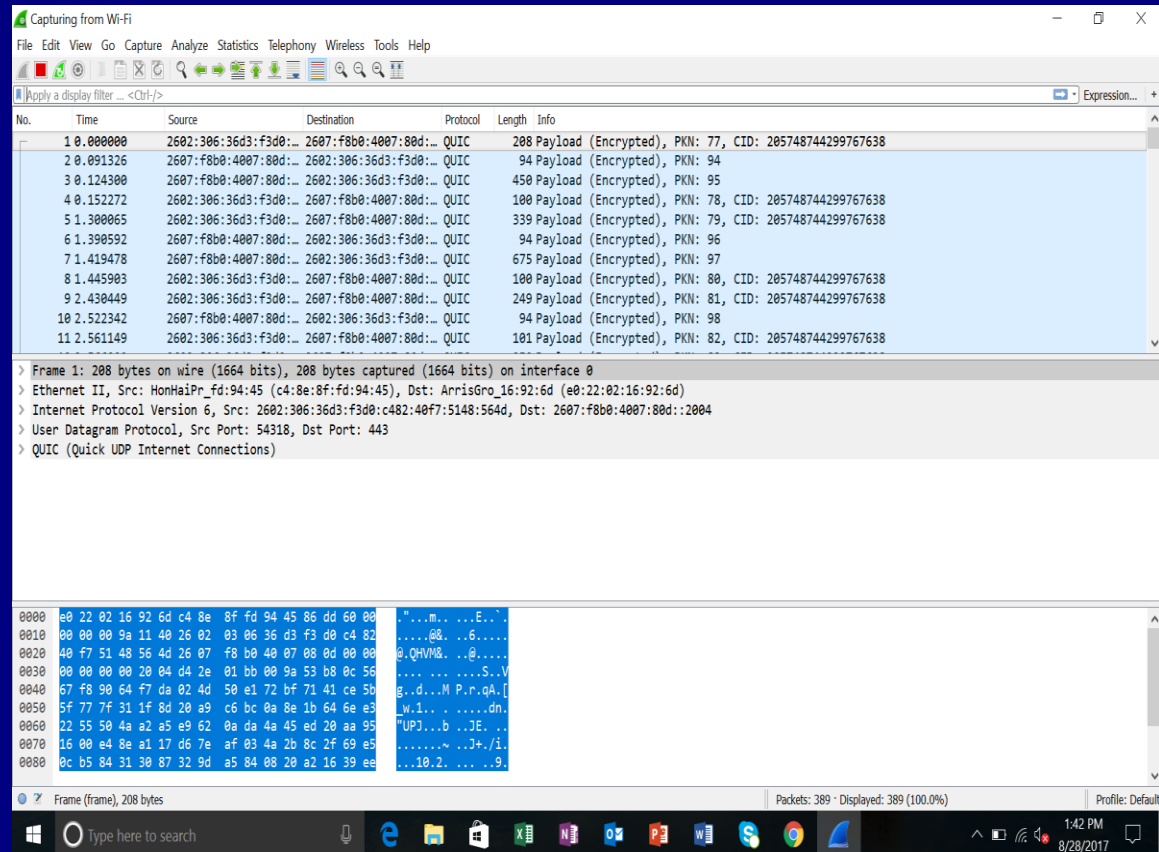
Once the capturing starts, main window will be blank until the data is exchanged on network interface (NIC)



Wireshark Usage

Capturing₂

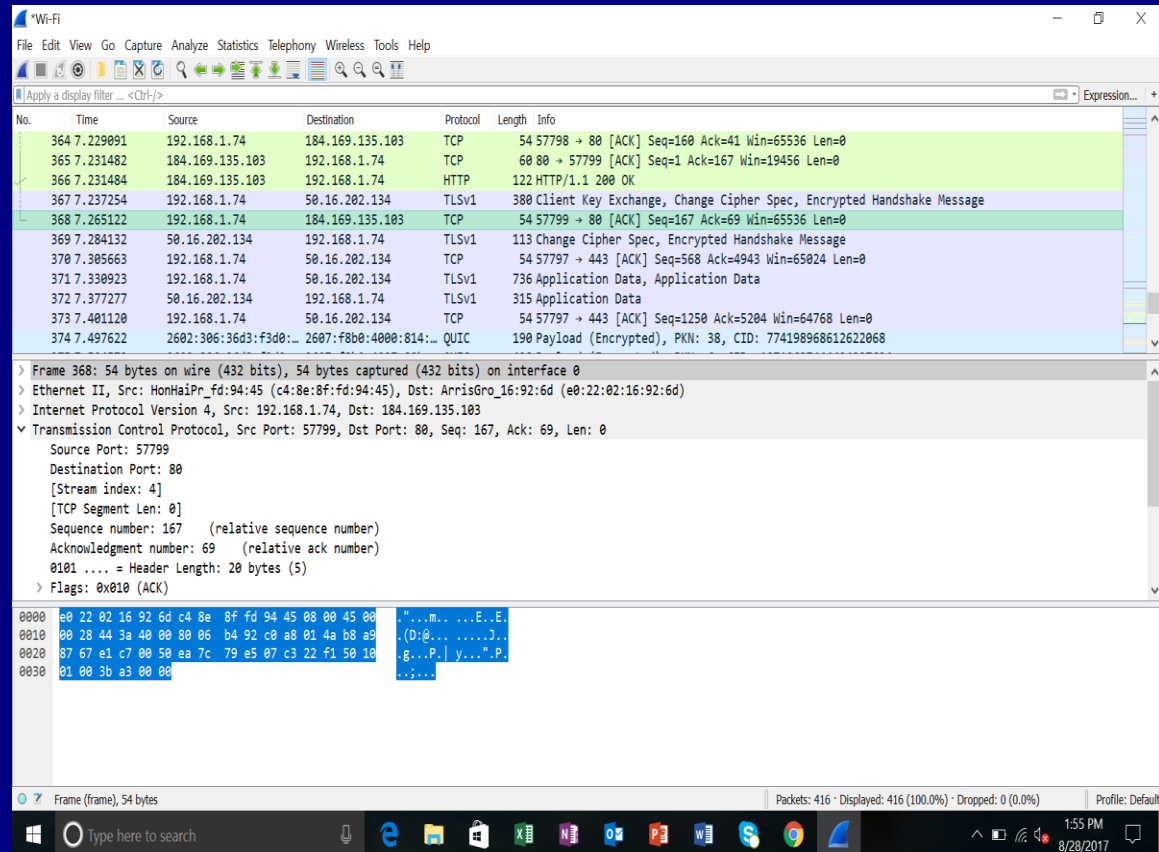
When packets exchanged on NIC, the packets will be dumped to main window



Wireshark Usage

Stopping Capture

Capturing can be stopped by clicking on Stop the running capture button on the main toolbar

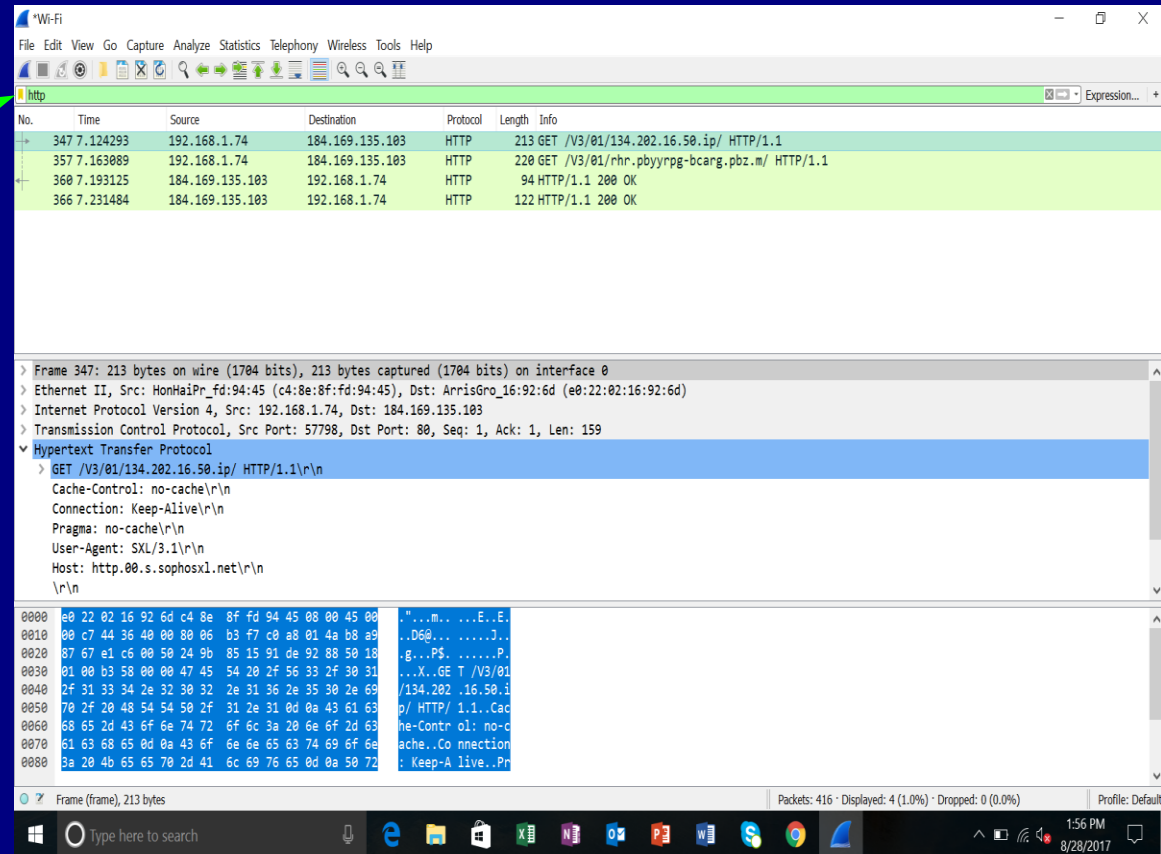


Wireshark Usage

Filtering

Filter by entering the protocol or field name in Apply a display filter and enter

Detailed filters can be applied by creating expressions



Protocol Analysis with Wireshark

Protocol Analysis and Examples

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Protocol Analysis with Wireshark

Protocol Analysis

- Packets and protocols can be analysed after capture
- Individual fields in protocols can be easily seen
- Graphs and flow diagrams can be helpful in analysis

Protocol Analysis and Examples

Packet Details Pane

Analysis is performed manually

Example shows TCP segment with SYN and ACK fields set to 1

The image shows a Wireshark packet capture window titled '*Wi-Fi'. The packet list pane shows three packets. The selected packet is packet 356, a TCP segment from 192.168.1.74 to 184.169.135.103, sequence number 54 57799, length 80. The packet details pane shows the following information:

- Transmission Control Protocol, Src Port: 80, Dst Port: 57799, Seq: 0, Ack: 1, Len: 0
- Source Port: 80
- Destination Port: 57799
- [Stream index: 4]
- [TCP Segment Len: 0]
- Sequence number: 0 (relative sequence number)
- Acknowledgment number: 1 (relative ack number)
- 1000 = Header Length: 32 bytes (8)
- Flags: 0x012 (SYN, ACK)
 - 000. = Reserved: Not set
 - ...0. = Nonce: Not set
 - 0... = Congestion Window Reduced (CWR): Not set
 - 0... = ECN-Echo: Not set
 - 0... = Urgent: Not set
 - 1... = Acknowledgment: Set
 - 0... = Push: Not set
 - 0... = Reset: Not set
 - 1... = Syn: Set
 - 0... = Fin: Not set
 - [TCP Flags:A..S.]
- Window size value: 17922
- [Calculated window size: 17922]
- Checksum: 0x9968 [unverified]
- [Checksum Status: Unverified]
- Urgent pointer: 0
- Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted, No-Operation (NOP), Window scale
- [SEQ/ACK analysis]

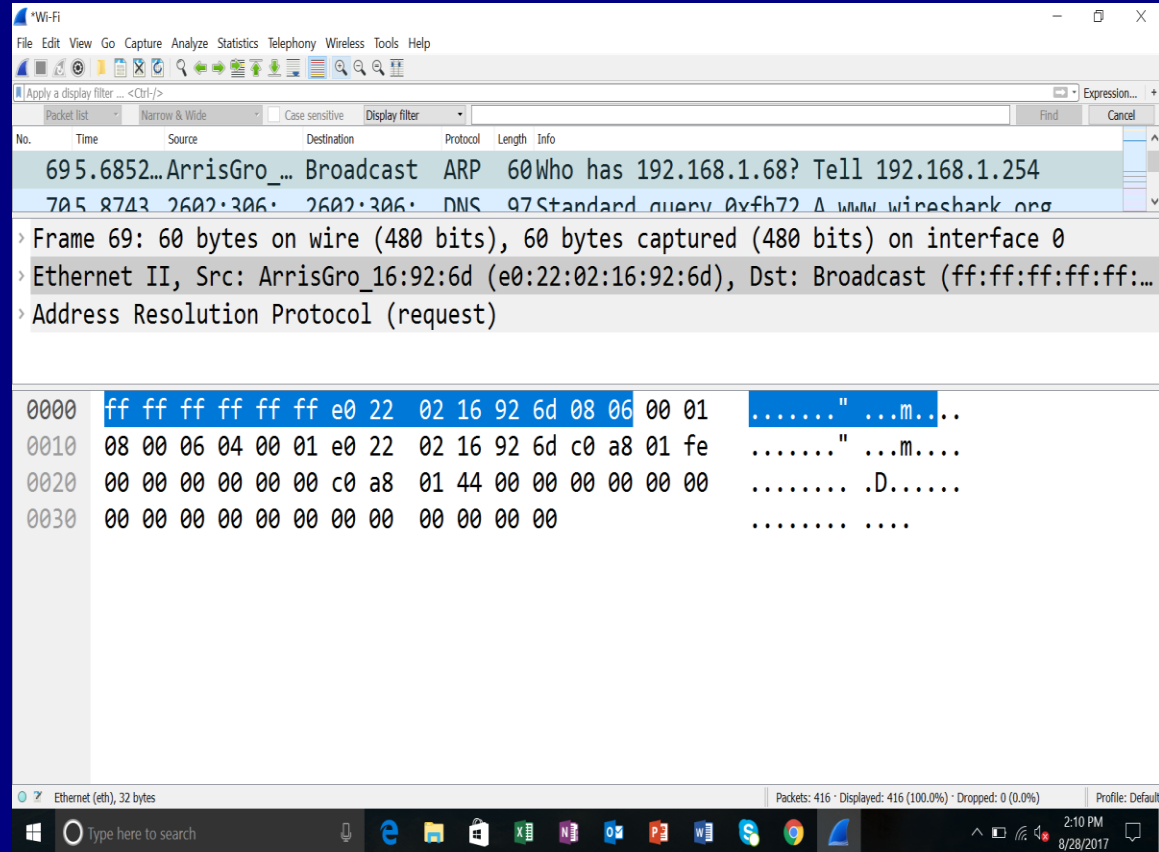
The packet bytes pane shows the raw data for the packet, including the Ethernet II header and the IP header.

Protocol Analysis and Examples

Packet Byte Pane

Zoom in or out is possible in main toolbar

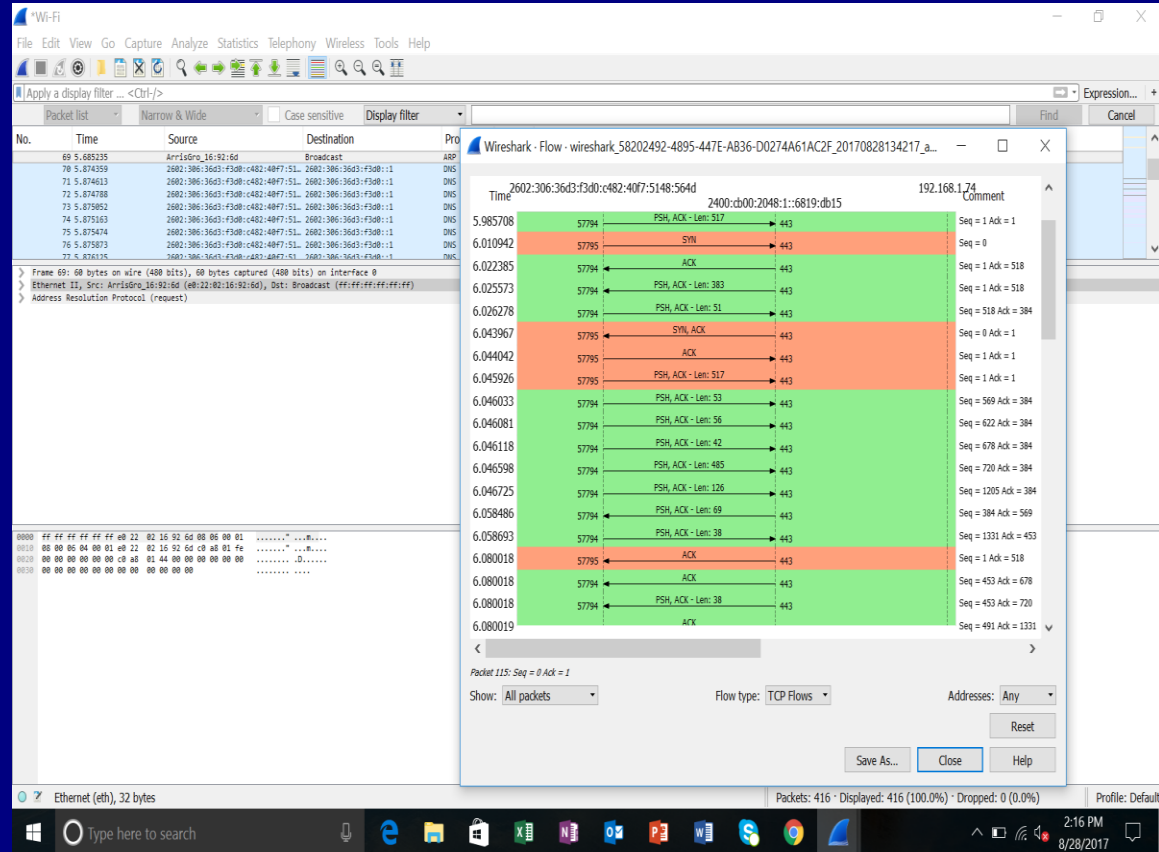
Packet Byte pane consists of offset, Hex, and ASCII fields



Protocol Analysis and Examples

Statistics – Flow Graph Example

TCP plots and flow graphs are available in Statistics menu



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Getting Started

Installation and First Lab Exercise

- Install Wireshark and familiarise
- Download first **Getting Started v7.0** exercise
<http://www-net.cs.umass.edu/wireshark-labs/>
- Complete first Wireshark Lab – *Getting Started*

Getting Started

First Lab Exercise Submission

- Complete first Wireshark Lab – *Getting Started*
 - follow EECS 780 submission instructions and email report
 - to grader cc to professor
 - Subject: EECE780 - wireshark getting started
 - attach file *<lastname>-wireshark-getting-started.pdf*

Protocol Analysis with Wireshark

Acknowledgements

Some material in these foils comes from the textbook
supplementary materials:

- Kurose & Ross,
*Computer Networking:
A Top-Down Approach*, 7th ed.
<http://kuroseross.com>
- <http://www.wireshark.org/>
- <http://www.winpcap.org/>